OmniAssist — Web/PWA Tasks

> Ordered work breakdown derived from `SPECIFICATION.md` and `IMPLEMENTATION.md`. Each task should fit one focused coding session.

Summary

Metric	Value
Tasks	24
Phases	7
Foundation	Task 4
Core CRUD	Task 10
Full MVP	Task 20
Production release	Task 24

Phase 1: Foundation

Task 1: Web-first repository migration

Files: `apps/web/**`, `apps/client-flutter-archive/**`, `Makefile`, `README.md`, `.gitignore`, `pnpm-workspace.yaml`.

Archive Flutter. Create Next.js 16 App Router app with TypeScript, Tailwind 4, strict linting, and lockfile. Keep `apps/api` separate. Update commands to target web/API.

Acceptance: `pnpm lint`, `pnpm typecheck`, and `pnpm build` pass; Flutter is outside active build; `make check` runs web/API checks.

Dependencies: None. **Effort:** 2–4 hours.

Task 2: API configuration and deployment baseline

Files: `apps/api/src/config.py`, `apps/api/src/main.py`, `apps/api/src/api/error_handlers.py`, `infra/render.yaml`, `infra/env/.env.example`, `apps/api/Dockerfile`.

Add typed settings, request IDs, safe errors, CORS, liveness/readiness, Render start command, and production env documentation.

Acceptance: API starts with `uv run uvicorn`; health routes return 200; production settings reject development credentials; Render blueprint validates.

Dependencies: Task 1. **Effort:** 3–5 hours.

Task 3: PWA shell and design tokens

Files: `apps/web/src/app/layout.tsx`, `globals.css`, `manifest.ts` or `public/manifest.webmanifest`, `public/sw.js`, `public/offline.html`, `public/icons/*`, `src/app/service-worker-registration.tsx`.

Implement the OmniAssist brand tokens, responsive shell, manifest, service worker, offline fallback, metadata, icons, and reduced-motion behavior.

Acceptance: Lighthouse/PWA checks find manifest and service worker; app shell renders at 320 px and desktop width; no private API responses are cached.

Dependencies: Task 1. **Effort:** 3–5 hours.

Task 4: Contracts, shared types, and CI

Files: `packages/contracts/*`, `apps/web/src/lib/types/*`, `apps/web/src/lib/api/*`, `.github/workflows/web.yml`, `.github/workflows/api.yml`.

Define success/error envelopes, stable error codes, domain TypeScript types, API client helper, and CI checks.

Acceptance: TypeScript imports compile without `any`; API errors preserve request IDs; CI runs lint, typecheck, tests, API tests, and builds.

Dependencies: Tasks 1–3. **Effort:** 4–6 hours.

Phase 2: Database and API Core

Task 5: Neon schema and migrations

Files: `apps/api/src/infrastructure/db/*`, `apps/api/migrations/env.py`, `apps/api/migrations/versions/0001_core.py`, `apps/api/tests/integration/test_migrations.py`.

Create SQLAlchemy models/migration for users, workspaces, projects, tasks, time blocks, milestones, reviews, analyses, experiments, notifications, activities, exports, and preferences. Scope user-owned data by workspace.

Acceptance: Empty PostgreSQL migrates cleanly; constraints/indexes exist; downgrade test succeeds in a disposable database.

Dependencies: Task 2. **Effort:** 6–8 hours.

Task 6: Auth claims and workspace authorization

Files: `apps/api/src/auth/*`, `apps/api/src/api/dependencies.py`, `apps/api/src/domain/users/*`, `apps/api/src/domain/workspaces/*`, tests.

Verify managed-auth JWT issuer/audience/signature/expiry, map provider subject to local user, resolve personal workspace, and reject cross-workspace access.

Acceptance: Valid token resolves user; invalid token returns 401; cross-workspace IDs return safe 403/404; no password routes exist.

Dependencies: Tasks 2, 5. **Effort:** 5–7 hours.

Task 7: Project and task API

Files: `apps/api/src/domain/projects/*`, `apps/api/src/domain/tasks/*`, `apps/api/src/api/routes/projects.py`, `tasks.py`, tests.

Implement CRUD, validation, statuses, tags, subtasks, due dates, recurrence metadata, completion timestamps, archive/delete behavior, pagination, and scoped repositories.

Acceptance: Valid CRUD works; invalid status/empty title fails; cross-workspace access fails; list endpoints paginate.

Dependencies: Tasks 5–6. **Effort:** 6–8 hours.

Task 8: Calendar, reminders, and milestones API

Files: `apps/api/src/domain/time_blocks/*`, `milestones/*`, `notifications/*`, routes, tests.

Implement blocks, overlap warnings, milestone progress/dependencies, reminder rules, quiet hours, and notification history.

Acceptance: Date validation works; overlaps warn without blocking; dependency cycles fail; completed tasks suppress future reminders.

Dependencies: Task 7. **Effort:** 7–8 hours.

Task 9: Research modules API

Files: `apps/api/src/domain/reviews/*`, `analyses/*`, `experiments/*`, routes, tests.

Implement configurable review stages/items, CSV validation/import/export, analysis ledger, experiment templates, finalization, revisions, and safe content handling.

Acceptance: Review stage movement creates history; invalid CSV reports rows; finalized experiment is immutable; revisions preserve prior content; no arbitrary code executes.

Dependencies: Tasks 7–8. **Effort:** 8 hours.

Task 10: Search and export API

Files: `apps/api/src/domain/search/*`, `exports/*`, routes, workers, tests.

Implement authorized search/filtering, JSON/CSV export jobs, status polling, expiring downloads, and account deletion workflow.

Acceptance: Search scopes records; export preserves relationships; export failure is recoverable; deletion revokes sessions and removes user data according to policy.

Dependencies: Tasks 6–9. **Effort:** 6–8 hours.

Phase 3: Authenticated Frontend

Task 11: Clerk integration and protected routes

Files: `apps/web/src/middleware.ts`, `src/app/(auth)/*`, `src/app/(app)/app/layout.tsx`, `src/lib/auth/*`, package files.

Add managed auth SDK, sign-in/up/reset screens, route protection, API token attachment, and signed-out states.

Acceptance: Signed-out user cannot access `/app`; signed-in user reaches Today; token is never logged; auth callback URLs are documented.

Dependencies: Task 6. **Effort:** 4–6 hours.

Task 12: Web application shell

Files: `apps/web/src/components/app-shell/*`, `src/app/(app)/app/layout.tsx`, `src/app/(app)/app/today/page.tsx`, tests.

Build responsive sidebar/mobile navigation, page header, sync/connection state, loading/error/empty states, and accessible landmarks.

Acceptance: Keyboard navigation works; mobile and desktop layouts are usable; no route uses placeholder starter UI.

Dependencies: Tasks 3, 4, 11. **Effort:** 5–7 hours.

Task 13: Onboarding and workspace UI

Files: `apps/web/src/app/(app)/onboarding/*`, `src/components/features/workspaces/*`, API actions, tests.

Implement timezone, working hours, preferences, initial workspace/project creation, skip/resume behavior, and form draft preservation.

Acceptance: New user completes or skips onboarding; invalid fields remain editable; first project can be created; refresh does not lose draft.

Dependencies: Tasks 6, 11–12. **Effort:** 5–7 hours.

Task 14: Projects and tasks UI

Files: `apps/web/src/app/(app)/app/projects/*`, `inbox/*`, `components/features/tasks/*`, `lib/features/tasks/*`, tests.

Build project list/detail, quick capture, Inbox, task create/edit/complete/archive, tags, subtasks, and mutation error recovery.

Acceptance: User creates a project and task end-to-end; field errors display; failed write preserves form; duplicate submit is prevented.

Dependencies: Tasks 7, 12–13. **Effort:** 8 hours.

Task 15: Today and calendar UI

Files: `today/*`, `calendar/*`, time-block components, tests.

Build daily briefing, agenda/week/month views, time-block editor, overlap warnings, rescheduling controls, and responsive mobile behavior.

Acceptance: Task and block appear in Today/calendar; overlap warning does not block save; drag has button alternative; mobile tap-to-edit works.

Dependencies: Tasks 8, 14. **Effort:** 8 hours.

Task 16: Milestone UI

Files: `projects/[id]/milestones/*`, milestone components, tests.

Build progress dashboard, timeline, dependencies, at-risk/blocked states, and create-linked-task action.

Acceptance: Manual/weighted progress displays correctly; cycle error is understandable; linked task opens with project/milestone prefilled.

Dependencies: Tasks 8, 14–15. **Effort:** 6–8 hours.

Phase 4: Research Frontend

Task 17: Review board and CSV flows

Files: `projects/[id]/reviews/*`, review components, CSV import/export client helpers, tests.

Build desktop board, mobile grouped list, stage editor, item detail, import preview/mapping/errors, and export request UI.

Acceptance: Item moves between stages; stage history visible; invalid CSV rows shown before commit; mobile flow works without drag-and-drop.

Dependencies: Tasks 9, 12, 14. **Effort:** 8 hours.

Task 18: Analysis ledger UI

Files: `projects/[id]/analysis/*`, analysis components, tests.

Build ledger/detail/editor, status transitions, environment/repository fields, result/error display, and next-action task creation.

Acceptance: Record saves and reloads; invalid transitions fail; next action creates linked task; code is never executed.

Dependencies: Tasks 9, 14, 16. **Effort:** 6–8 hours.

Task 19: Experiment log UI

Files: `projects/[id]/experiments/*`, experiment components, tests.

Build templates, draft autosave, finalization, revision view, warning copy, and conflict-safe update behavior for online-first writes.

Acceptance: Draft survives reload; finalized record cannot be silently edited; revision preserves prior version; warning is visible.

Dependencies: Tasks 9, 14, 16. **Effort:** 8 hours.

Task 20: Search, notifications, export, settings

Files: `search/*`, `notifications/*`, `settings/*`, shared components, tests.

Build global search/filter, notification center/preferences, export status/download, account deletion confirmation, and cache cleanup.

Acceptance: Search results are authorized; export can be downloaded; deletion requires explicit confirmation; user can recover from failed export.

Dependencies: Tasks 10, 15–19. **Effort:** 8 hours.

Phase 5: Quality and PWA

Task 21: Browser persistence and PWA verification

Files: `src/lib/drafts/*`, service worker, Playwright tests, manifest/icons.

Bound draft persistence, add offline shell tests, verify Android Chrome and iOS Safari install instructions, and ensure private API data is not cached.

Acceptance: Shell loads offline; drafts recover; API cache does not contain auth/private data; install checklist passes on target browsers.

Dependencies: Tasks 3, 12–20. **Effort:** 5–7 hours.

Task 22: Accessibility and responsive QA

Files: shared UI and feature components, accessibility tests, `docs/accessibility-checklist.md`.

Audit focus, semantics, contrast, touch targets, keyboard alternatives, reduced motion, and 320 px–desktop layouts.

Acceptance: Critical paths pass automated accessibility checks; keyboard-only flow reaches all actions; status never relies on color alone.

Dependencies: Tasks 12–20. **Effort:** 6–8 hours.

Task 23: Security, performance, and integration QA

Files: auth/API middleware, security tests, Playwright flows, performance notes.

Run authorization matrix, input/file limits, rate limits, response timing, error redaction, and full web-to-API E2E tests.

Acceptance: Cross-workspace tests pass; private content absent from logs; p95 targets are measured; core journey passes on mobile viewport.

Dependencies: Tasks 6–21. **Effort:** 8 hours.

Phase 6: Deployment

Task 24: Production release

Files: `.github/workflows/*`, `infra/render.yaml`, `infra/neon.md`, `README.md`, `docs/deployment.md`, `docs/known-limitations.md`.

Configure Vercel, Render, Neon, managed auth, migrations, CORS, domain, monitoring, backups, health checks, release notes, and rollback steps.

Acceptance: Production web build deploys; Render health check passes; Neon migration succeeds; auth callback works; Android/iOS PWA install and create-task journey work; rollback is documented.

Dependencies: Tasks 1–23. **Effort:** 6–8 hours.

Milestones

Milestone	After	Result
Foundation	4	Web shell and API checks pass
Data layer	10	Authenticated CRUD API works
Core product	16	Planning workflow works
Research MVP	20	Research modules and settings work
Quality	23	Security, accessibility, PWA, and E2E gates pass
Release	24	Vercel/Render/Neon deployment is usable

Dependency graph

Plain Text

```
T1 → T2 → T3 → T4 → T5 → T6 → T7 → T8 → T9 → T10
              ↘ T11 → T12 → T13 → T14 → T15 → T16
                                          ↘ T17 → T18 → T19 → T20
T3 + T12–T20 → T21 → T22 → T23 → T24
```